METRIC
MIL-DTL-23443C
10 December 2019
SUPERSEDING
DOD-D-23443B
15 May 1978

DETAIL SPECIFICATION

DI-(2-ETHYLHEXYL) ADIPATE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of di-(2-ethylhexyl) adipate, technical, herein after referred to as the plasticizer.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D92	-	Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D1045	-	Standard Test Methods for Sampling and Testing Plasticizers Used in Plastics
ASTM D1209	-	Standard Test Method for Color of Clear Liquids (Platinum-Cobalt Scale)
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM E203	-	Standard Test Method for Water Using Volumetric Karl Fischer Titration

(Copies of these documents are available online at www.astm.org.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:54Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4.

3.2 Color and clarity. The color shall be no darker than the color standard prepared in accordance with 4.6.1. The plasticizer shall be a clear liquid containing no visible suspended solid matter when examined in accordance with 4.6.2.

3.3 Physical and chemical properties. The physical and chemical properties of the plasticizer shall be in accordance with table I.

TABLE I. Physical and chemical properties.

Property	Requirement		Test Procedure
	Min	Max	
Specific gravity, 25/25 °C	0.921	0.929	4.6.3
Moisture, wt % 1/	--	0.05	4.6.4
Acidity, wt % 1/	--	0.01	4.6.5
Saponification number	300	304	4.6.6
Flash point, °C	196	--	4.6.7
Fire point, °C	216	--	4.6.7
Kinematic viscosity, 38 °C, mm^2/s (see 6.5)	8.00	8.76	4.6.8

NOTE:
1/ Percentage by weight.

3.4 Workmanship. The material shall be uniform and free from contamination or any other defect that would prevent its use for the purpose intended when tested as specified in 4.6.2. The material shall be manufactured in accordance with standard manufacturing procedures of the industry.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Conformance inspection (see 4.5).

4.2 Inspection conditions. Unless otherwise specified in the contract (see 6.2), inspection conditions shall be room ambient conditions.

4.3 Sampling.

4.3.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all plasticizer manufactured in one continuous production run or in one batch, under essentially identical conditions, from the same raw materials, and offered for acceptance at one time. Several batches manufactured from the same raw materials may be blended to form a uniform larger batch which shall then constitute a lot for inspection purposes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:54Z
Check the source to verify that this is the current version before use.

4.3.2 Sampling for conformance inspection. Samples for conformance inspection shall be obtained in accordance with the following:

4.3.2.1 Containers. The number of containers to be sampled shall be selected in accordance with the minimum number of packages per lot specified in ASTM D4057. Primary samples shall be obtained from the sample containers in accordance with the tube sampling method of ASTM D4057. Each primary sample shall consist of approximately 0.5 liters (L) of plasticizer. A composite sample of approximately 1.0 liter shall be prepared from equal portions of each primary sample, placed in airtight glass bottles, and labeled with the material name, manufacturer, plant, contract and lot numbers, and lot size. All primary sample remnants shall be held for possible future reference.

4.3.2.2 Tank cars. Each tank car shall be sampled in accordance with the bottle or beaker method of ASTM D4057. The sample shall consist of approximately 1.0 liter of plasticizer and shall be placed in an airtight glass bottle labeled as specified in 4.3.2.1.

4.4 First article inspection. When required (see 6.2), the first article inspection shall consist of all the tests required for conformance inspection (see 4.5) in addition to any additional inspections specified (see 6.2).

4.5 Conformance inspection. Unless otherwise specified in the contract (see 6.2), the conformance inspection shall consist of the tests and examinations detailed in 4.6 performed on samples selected in accordance with 4.3.2. Failure of the composite sample to pass any test or examination shall result in rejection of the lot.

4.6 Tests and examinations. Unless otherwise specified herein, all chemicals shall be American Chemical Society or Reagent grade. The material shall be tested in accordance with the following test methods. Alternate material test methods may be used if approved in writing by the Technical Authority (see 6.7 and 6.9).

4.6.1 Color. The color shall be tested as follows:

a. Prepare a No. 50 Platinum-Cobalt Color Standard in a 100-milliliter Nessler tube as described in ASTM D1209 (10 milliliters No. 500 Platinum-Cobalt Stock Solution diluted to 100 milliliters). The No. 500 Platinum-Cobalt Stock Solution, already prepared, is available from laboratory supply houses. It may be used provided it meets the absorbance tolerance limits specified in ASTM D1209 for the No. 500 Platinum-Cobalt Stock Solution.

b. Fill a clean 100-milliliter Nessler tube to the mark with the plasticizer sample.

c. Hold the sample and standard tubes side by side, and compare their colors using transmitted light while looking downward through the tubes.

d. The sample shall be rejected if the color is darker than that of the No. 50 Platinum-Cobalt Standard Solution.

4.6.2 Clarity and workmanship. The clarity and workmanship shall be examined by inverting a bottle of the sample several times and visually examining the plasticizer. The presence of any suspended solid material shall result in rejection of the sample.

4.6.3 Specific gravity. The specific gravity of the plasticizer shall be determined by using the pycnometer method in ASTM D1045, except the plasticizer sample shall be cooled to below 25 °C and the specific gravity shall be determined at 25/25 °C instead of 23/23 °C.

4.6.4 Moisture. The moisture content shall be determined in accordance with ASTM E203 using anhydrous methanol as the titration solvent. The sample size shall be 10 milliliters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:54Z
Check the source to verify that this is the current version before use.

4.6.5 Acidity. Two drops of 1-percent phenolphthalein indicator shall be added to 50 milliliters of absolute ethanol in a 125-milliliter flask. The resulting solution shall be titrated with standardized 0.1 normal sodium hydroxide until a faint pink color can be observed in the flask. This color shall remain pink for not less than 30 seconds. The weight of the flask and its contents shall be determined to the nearest 0.1 gram. About 25 milliliters of sample plasticizer shall be added and the flask shall be reweighed to the nearest 0.1 gram to determine how much of the material was added to the flask. The solution shall be titrated with standardized 0.1 normal sodium hydroxide solution until a faint pink color can be observed in the flask. This color shall remain for at least 30 seconds. If the titration exceeds 10 milliliters, repeat the determination using standardized 0.1 normal sodium hydroxide. The percent acetic acid shall be calculated as follows:

$$\text{Percent acetic acid} = \frac{6.0\ VN}{W}$$

Where:

V = Volume of sodium hydroxide to titrate sample solution, milliliters

N = Normality of sodium hydroxide

W = Weight of sample, gram

4.6.6 Saponification number. The saponification number shall be determined in accordance with the titrimetric ester content procedure in ASTM D1045 using a 3-hour reflux time except: 10 drops of 1 percent phenolphthalein indicator solution shall be used in lieu of bromothymol blue indicator; the 0.5 N potassium hydroxide need not be standardized; 24-inch air condensers with ground glass joints may be used in place of water-cooled condensers; a 50-milliliter Class A burette with 0.1-milliliter graduations shall be used to deliver the hydrochloric acid (HCl); and the glassware must be chemically clean. It is recommended that the normality of the HCl, standardized to four decimal places, be slightly higher than 0.50 N (but not greater than 0.52 N) so the burette will not require refilling during titration of the blank. The saponification number calculation requires subtraction of the acid number of the sample. To convert the percent acidity (determined in 4.6.5) to acid number, multiply by 9.35. The saponification number shall be reported to the nearest whole number. Duplicate results should be considered suspect if they differ by more than three saponification numbers.

4.6.7 Flash point and fire point. The flash point and the fire point shall be determined in accordance with ASTM D92.

4.6.8 Kinematic viscosity. The kinematic viscosity in square millimeters per second (mm^2/s) (see 6.5) at 38 °C shall be determined in accordance with ASTM D445 using an Ubbelohde viscometer, Cannon-Fenske viscometer, or equal.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Di-(2-ethylhexyl) adipate in accordance with this specification is intended to be used as a plasticizer in propellant and explosive compositions. Successful use of the plasticizer in propellant and explosive formulations is dependent on specific properties of the plasticizer which are requirements this specification. Current stock of di-(2-ethylhexyl) adipate may be used until depleted; future requisitions for di-(2-ethylhexyl) adipate should refer to the current version of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:54Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity required.

c. When a first article is required (see 3.1 and 4.4).

d. Inspection conditions if other than as specified (see 4.2).

e. Lot size if other than as specified (see 4.3.1).

f. When additional first article inspections are required (see 4.4).

g. Conformance inspection plan if other than as specified (see 4.5).

h. Packaging requirements if other than specified (see 5.1, 6.3, and 6.3.1).

i. Size of container required (see 6.3.1).

j. Any special marking required (see 6.4).

k. Safety data sheet (SDS) requirements (see 6.6).

l. Any data required for delivery (use DD Form 1423).

6.3 Packaging and packing. Unless otherwise specified (see 6.2), packaging and packing should be level C.

6.3.1 Level C. Unless otherwise specified (see 6.2), packaging and packing of the plasticizer should be in accordance with method 10 of MIL-STD-2073-1. The packaging and packing should be of such construction and materials that the contents will be adequately protected against loss or contamination. Container size should be as specified (see 6.2). Containers should conform to Uniform Freight Classification, National Motor Freight Classification or to rules of other carriers applicable to the mode of transportation.

6.4 Marking. In addition to any special marking specified (see 6.2), each container should be marked in accordance with MIL-STD-129. Marking should include but not be limited to, the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and location.

c. Material trade name (see 6.8).

d. Net weight and volume.

e. Lot number, batch number(s), and date of manufacturer.

f. Storage conditions.

g. Toxicity precautions, if applicable.

h. Contract number.

6.5 Definition. The International System of Units (SI) recognizes the m^2/s as the official metric unit for kinematic viscosity. One centistoke (cSt) is therefore equal to one mm^2/s.

6.6 Safety data. When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.7 Alternate test methods. Other material test methods may be used providing that the proposed method is equivalent (in accuracy and precision) to the method given in this specification. Prior approval of NAVSEA is required for use of equivalent material test methods.

6.8 Trade name. Di-(2-ethylhexyl) adipate (CAS Number 103-23-1) is often called Dioctyl adipate or DOA by manufacturers. However, Dioctyl adipate (DOA) (CAS Number 123-79-5) is a constitutional isomer of di-(2-ethylhexyl) adipate and has different properties. The military specification for Dioctyl adipate is MIL-D-85494. Check the product technical data sheet or SDS to verify that the correct material is being used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:54Z
Check the source to verify that this is the current version before use.

6.9 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.10 Subject term (key word) listing.

Binder

Dioctyl adipate

DOA

Explosive

Plasticizer

Propellant

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – AR
Navy – OS
Air Force – 68

Review activities:
Army – MI
DLA – GS

Preparing activity:
Navy – OS
(Project 6810-2019-005)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:54Z
Check the source to verify that this is the current version before use.